

Towards Debiasing Movie Recommender System using Edge-dropout

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Abstract—Recommender systems are widely used across digital platforms, but they frequently inherit biases from training data, resulting in unfair recommendations for certain sociological groups. Most techniques for graph based recommender systems rely on Graph Neural Networks applied to a heterogeneous bi-partite graphs. Existing solutions for bipartite graphs to mitigate biases have focused on in-processing techniques due to negligible effect on graph topology. This work introduces a novel pre-processing technique for bipartite graphs based on a simple edge-dropout scheme adapted from FairDrop. The method involves dropping edges corresponding to the top dyadic groups containing sensitive attributes with a biased random probability similar to flip of a biased coin. Experiments with the MovieLens 100k dataset using GAT and GCN architectures, demonstrate that edge dropout can improve dyadic Statistical Parity and Equality of Opportunity difference fairness metrics with minimal trade-off to accuracy. Our findings reveal that GAT and GCN exhibit opposite behaviors under edge-based dropout suggesting that existing group fairness metrics are imperfect for movie recommender systems due to masking of sub-group interactions of gender with movie genres. When blindly utilizing equality and parity metrics this may exacerbate bias towards or remove preferences of certain user-groups which is undesirable.

Index Terms—GNN, edge-dropout, movie recommender system, dyadic fairness metrics

I. INTRODUCTION

Recommender systems have become essential to modern digital platforms, ranging from e-commerce websites to content streaming services. By leveraging vast amounts of user data, these systems provide personalized suggestions, significantly improving the user experience.

Despite their effectiveness, recommender systems often inherit biases from the data they are trained on. This is a common issue when the training population differs significantly from the target population, either due to inadequate sampling or demographic changes over time. For example, research has shown that IMDb, a leading movie review platform, receives a disproportionately large number of ratings from male users [19]. Biases in recommender systems [7] are problematic for two key reasons: First, they result in suboptimal recommendations that fail to serve a diverse range of users. Second, these systems can perpetuate social inequalities by reinforcing biases related to race, gender, or other sensitive attributes, raising ethical concerns for the platforms involved.

At the core of many recommendation systems is a Graph Neural Network (GNN) architecture, which is applied to a bipartite graph connecting users to items.

Many recommender systems use a type of graph that connects two groups of things, like users and the movies they watch. Think of it as a map where one side is users, the other side is movies, and the connections show who liked or watched what. This is called a *bipartite graph*. This is something we will utilize in this work for fair movie recommendations. Graph Neural Networks (GNNs) [20] are a type of neural network architecture that are often used for such type of graphs. A strong motivation for using GNNs compared to other neural network architectures as Deep Neural Networks is that GNNs assume that there is a relationship between data-points hence capturing structural properties of a network while also learning representations through matrix operations. These properties are something absent in DNNs.

While much research has focused on debiasing GNNs [5], there has been comparatively less attention on addressing bias in bipartite graphs. Most existing work on this topic has focused on in-processing and post-processing techniques [13], while pre-processing methods remain underexplored. In-processing refers to making changes to the neural network architectures, post-processing refers to keeping the architecture constant but only changing the prediction probabilities and finally pre-processing refers to the stage where we manipulate our dataset to be used for training our model such as a GNN.

In this paper, we aim to bridge this gap by exploring several pre-processing techniques for debiasing bipartite graphs. Inspired by prior work on *planar graphs* [17], we design a similar *edge sampling* technique tailored for *bipartite heterogeneous* graphs. FairDrop is one of those research techniques that can be applied to most type of graphs. This makes it a strong motivation to be utilized. The only limitation of FairDrop is that it doesn't work with planar graphs. Planar graphs don't have two groups connected to each other. Only a single group with nodes connected to each other. In our case we do not have such a graph therefore it allows us to make some contributions.

Our key contributions are:

- We introduce a novel edge sampling method, inspired from FairDrop [17], designed to improve fairness in heterogeneous bipartite graphs.

- We evaluate our method alongside standard pre- and post-processing techniques on the *MovieLens 100k* dataset for the link prediction task
- We discuss the results and motivate that fairness metrics may hide group differences across GNN architectures, underscoring the need for refined evaluation in graph-based recommendations.

II. BACKGROUND

Graph Neural Networks have become the State-of-The-Art architectures of modern recommender systems [21]. An extensive number of Graph Neural Networks architectures have been developed [20].

The definition of fairness itself is broad and optimizing for all types of fairness metrics is impossible[8]. Properly defining fairness usually entails optimizing for user utility as well as group or individuals using their respective sensitive attributes. In the absence of equal proportion of reported sensitive attributes for our users, counterfactual fairness approach can be used. However, assuming a causal model requires approximation in measurement of bias due to huge number of combinations of changed sensitive attributes. Fairness itself can be approached through a variety of methods from pre-processing, in-processing to post-processing methods. As shown by [7, 13] research into edge-based sampling methods mostly covers planar graphs. Very little work have been dedicated to debiasing bi-partite graphs [2, 3, 22].

III. METHOD

A. Dataset

We chose an bi-partite dataset for movie recommendations namely MovieLens 100k dataset to base our experiments. The dataset also contains an imbalance in the nodes of male and female users considered as sensitive users. The smaller version of the MovieLens dataset allow us explore the limits of pre-processing methods. In order to make the optimization problem for our GNN easier, the ratings from 1-5 were converted to Binary labels 0/1 where labels below 4 are converted to 0 and the rest as 1. Thus, we mask the complexity of training a recommendation system itself by minimizing the optimization problem to a simple classification problem. This simplifies our recommendation problem to a Link-Prediction task where a strong probability prediction or 1 would suggest high link formation hence a recommendation of a movie. Moreover, we can reuse metrics for training and testing for simple binary classification. A consequence of this modification is the output distribution of predicted labels being bi-modal in nature.

Number of users	943
Number of movies	1,682
Number of ratings	100,000
Number of 4&5 star ratings	55,375
Average number of 4&5 ratings by user	58.7
Average number of 4&5 ratings by movie	32.9

TABLE I: Statistics of the MovieLens 100k dataset.

Type of graph	Bipartite
Average shortest path	2.9
Diameter	6

TABLE II: Networks characteristics of the graph created from the MovieLens 100k Dataset.

B. Fairness Metrics

In my task of Movie recommendation, we look at the feature or category as gender. I looked at age groups before settling on gender, however, with age-groups we may run the risk of building a recommendation system that recommends adult movies to children and vice versa if the recommendation system is made fair. This would not be appropriate. Hence, the fairness would be across gender-groups. We have two groups, Male and Female. The dataset itself does not contain other gender categories which would be interesting to evaluate for fairness but absent nonetheless. In order to test our for group fairness we used Statistical Parity[1] and Equality Opportunity [9] as the set of fairness metrics.

Statistical Parity: In this metric we simply look if the probability of a prediction is same for each group we are evaluating for fairness. We want the difference of these probabilities to be somewhat equal or closer to zero.

$$\Delta DP = |P(\hat{Y} = 1|S = 0) - P(\hat{Y} = 1|S = 1)| \quad (1)$$

Equality of Opportunity: In this metric we look at if the probability of a prediction is same for each group given that a similar movie recommendation was present in the training dataset . We want the difference of these probabilities to be somewhat equal or closer to zero. similar label

$$\Delta EO = |P(\hat{Y} = 1|Y = 1, S = 0) - P(\hat{Y} = 1|Y = 1, S = 1)| \quad (2)$$

C. GNN Architectures

As motivated above on the usage of GNNs , we use a special type of GNN called the GCN. It is essentially adopted from a common Deep Neural Network architecture called Convolutional Neural Network. This is adapted from graphs in the form called Graph Convolutional Neural Networks (GCN). The manner in which this networks learns node(user and movie node) representations is by first aggregating representations of the neighboring nodes to which a single node is connected to which is called an aggregation. After aggregation which is usually done as a weighted sum, we add this aggregated information to the nodes own representation by addition called an update step. We usually do this sort of process for each layer, hence a GNN can have many layers similar to a DNN. The difference being that GNN representations encodes structural information through the aggregation procedure. In this given movie recommendation think of each user's node representation vector being an combination of other movies it is connected(rated) to. The equations of the same are explained below. But given the layer effect each user representation would also learn indirectly some information about its similar

users as they would watch similar movies! This is exactly where the dilemma of fairness occurs, due to this assumption that is used for training the GNN. The mathematical forms are explained below for the architectures we use below. We also use another variant of GCN that performs aggregation from neighbors but dynamically weights based on a learning algorithm.

We trained different GNN architectures such as GAT[18], GCN [10] using an 80%-20% split. We initially used GNN's with different layers and later we chose 2-layer only due to the short radius of 3 and average shortest path length of 2.9 as observed in Table II.

$$x'_i = \mathbf{W}_1 x_i + \mathbf{W}_2 \sum_{j \in \mathcal{N}(i)} e_{j,i} \cdot x_j$$

$$\mathbf{x}'_i = \alpha_{i,i} \Theta_s \mathbf{x}_i + \sum_{j \in \mathcal{N}(i)} \alpha_{i,j} \Theta_t \mathbf{x}_j$$

Both networks aggregate information from neighbouring nodes differently. While GAT aggregates based on a self-attention mechanism $\alpha_{i,i}$, GCN uses edge-weighting $e_{j,i}$ for aggregating representations. This we believe might play a crucial role in how both architectures behave with respect to fairness metrics.

Although it has been shown that bias exists in networks even when the sensitive features are removed due to the correlation between other features associated with sensitive features this can lower the accuracy. While conducting our experiments we noticed that including gender features leads to a marked improvement in overall accuracy a trait desirable for accurate recommendations. Hence, all the GNN's included gender as the sensitive attribute rather than removing it all together. We look at Accuracy, AUC, F1 as utility metrics.

D. Non-Fairness Evaluation Metrics

Accuracy metric is used for quantifying how well we are predicting suggested movie links. AUC measures the model's ability to distinguish between links and non-links across different decision thresholds. F1-Score metric balances the effect of capturing relevant links (Recall) and avoiding false positives (Precision). Its especially useful when dataset is imbalanced with respect to links/no-links. AUC measures the model's ability to distinguish between links and non-links across different decision thresholds. A higher AUC indicates better overall performance. We will mostly focus on Accuracy, F1-Score and AUC.

$$\text{Accuracy}(\text{Acc}) = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (3)$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

$$\text{Precision}(\text{Pr}) = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (5)$$

$$\text{Recall}(\text{Rec}) = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (6)$$

$$\begin{aligned} \text{True Positive Rate}(\text{TPR}) &= \frac{\text{TP}}{\text{TP} + \text{FN}}, \\ \text{False Positive Rate}(\text{FPR}) &= \frac{\text{FP}}{\text{FP} + \text{TN}} \end{aligned} \quad (7)$$

$$\text{AUC} = \sum_{i=1}^{n-1} \Delta \text{FPR}_i \cdot \frac{\text{TPR}_i + \text{TPR}_{i+1}}{2} \quad (8)$$

E. Dyadic Fairness Metrics

Using fairness metrics such as SP and EO which assume structure of data is i.i.d can mask the extent of bias itself. This is not desirable especially for bi-partite graph structures where a source and target representations are effected due to interactions between them, hence not i.i.d. Therefore, we adapt the usual fairness metrics to dyadic fairness metrics considering the combination of user- and movie-sensory attributes in our metric. Thus a dyad represent an specific set representing the combination of user-movie sensitive feature, hence an edge. This means instead of just looking at users or movies individually, we measure fairness by analyzing their interactions. Dyadic metrics, however, cannot be applied directly to bi-partite graph structures due to the absence of intra-links (links between same group, because this is a bipartite graph), that are responsible for the formation of "filter-bubbles". Of the three dyadic metrics proposed by [17] only Sub-Group dyads are applicable. Sub-Group dyads cannot be used as they would assume distributions across node-sets together hence assuming it as a planar graph. Mixed-dyads cannot be used as well as they require comparison across intra and inter groups of dyads, which is not possible in the current case. The fairness equations for the case where only two dyadic groups are presented below, which are then adopted to multiple groups as presented by [17]. In order to adopt our fairness metrics from i.i.d data to dyadic graphs data especially for bi-partite we can first replace S terms by dyads D instead. Dyadic groups here represent a unique combination of user and movie sensitive attributes. For our given case this would entail $S_U = \{\text{Age, Gender, Occupation}\}$ and $S_M = \{\text{Category}\}$ where $S_U \cap S_M = \emptyset$. In our given case we focus on dyads consisting of only Gender and Categories. Thus our dyads are represented as $D = \{\text{Age} \cap \text{Category}\}$. The Sub-Group dyadic distribution of edges is shown in Figure 1.

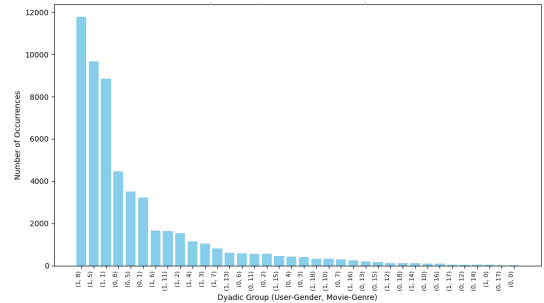


Fig. 1: Sub-Group Dyadic distribution of age-category edges shows male-dominated dyads.

Dyadic Statistical Parity

$$\Delta DP = \left| P(\hat{Y} = 1 | D = 0) - P(\hat{Y} = 1 | D = 1) \right| \quad (9)$$

Dyadic Equality of Opportunity

$$\Delta EO = \left| P(\hat{Y} = 1 | Y = 1, D = 0) - P(\hat{Y} = 1 | Y = 1, D = 1) \right| \quad (10)$$

Dyadic Equations for Multiple Sensitive Attributes

$$\Delta DP = \max_s \mathbb{E}[\hat{Y} | D = s] - \min_s \mathbb{E}[\hat{Y} | D = s] \quad (11)$$

$$\Delta TPR = \max_s \mathbb{E}[\hat{Y} = 1 | D = s, Y = 1] - \min_s \mathbb{E}[\hat{Y} = 1 | D = s, Y = 1] \quad (12)$$

$$\Delta FPR = \max_s \mathbb{E}[\hat{Y} = 1 | D = s, Y = 0] - \min_s \mathbb{E}[\hat{Y} = 1 | D = s, Y = 0] \quad (13)$$

$$\Delta EO = \max(\Delta TPR, \Delta FPR) \quad (13)$$

Node Representational Bias(RB)

In order to evaluate the amount of bias encoded within our user node embeddings we use

$$\text{Node RB} = \sum_{s \in S} \frac{|Z_s|}{|V|} \text{Accuracy}(\hat{S}, S) \quad (14)$$

here $\hat{S} = f(Z_s)$ represents the predicted gender sensitive attribute after training a classifier f on the retrieved user node embeddings Z as see in Algorithm 1. Some implementations often use *AUC* instead of Accuracy when the output is probabilities instead of binarized label outputs [2].

Algorithm 1 Calculating Node Representational Bias

Input: Embedding matrix X_B , sensitive labels Y_B

Output: Fairness results r_{dict}

- 1: Split X_B, Y_B into $(X_{train}, X_{test}, Y_{train}, Y_{test})$
 - 2: Classifiers $C \leftarrow \{\log, \text{mlp}, \text{rf}\}$ $\triangleright$ Specify classifiers for evaluation
 - 3: **for all** $c \in C$ **do**
 - 4: Train c on (X_{train}, Y_{train}) $\triangleright$ Train the classifier
 - 5: $c_{best} \leftarrow \arg \max_c \text{acc}(X_{test}, Y_{test})$ $\triangleright$ Select best current classifier based on cross-val accuracy
 - 6: Compute $\text{BalancedAccuracy}(c_{best})$
 - 7: $r_{dict} \leftarrow r_{dict} \cup \{\text{BalancedAccuracy}(c_{best})\}$
 - 8: **end for**
 - 9: **return** $\frac{1}{|r_{dict}|} \sum_{c \in r_{dict}} \text{BalancedAccuracy}(c)$
-

F. Pre-processing Methods

We consider pre-processing methods as the primary means of debiasing our network. Specifically we look at Edge Sampling based methods only, without changing the topology itself. Each method is adapted to dyads to account for dyadic fairness properties. In our methods we mostly deal with dyads as edges as it allows us to abstractly view in a single representation edges with sensitive attributes of source and target nodes. Pre-processing methods being the first stage of the training process is easy implement and almost akin

to augmentation on the dataset can be incorporated in data engineering pipelines as well. A variety of pre-processing methods were tested including Randomized Edge Drop and Fair(Biased) Edge Drops. This however is not straightforward as we deal with bi-partite graph structures and are observing dyads. To circumvent this we implement a dyadic masking strategy.

1) *Dyadic Masking*: Once we extract the distribution of the frequency of edges of all unique dyads, we can select the dyads in the 95th percentile as they cause the most bias due to a combination of popularity and gender imbalance. We label all edges in these dyads as 1 and rest as 0. This makes the process of dropping edges a more controlled task with respect to our sensitive attributes itself.

2) *Random Edge Dropouts via Dyadic Masking*: An simple method for developing our intuition of fair pre-processing is through randomly dropping edges in our graph. This however cannot be done in a manner that disrupts sparse dyadic edge population. Here edges are dropped by considered masked and unmasked dyads with equal probability. This method is in essence a good control to evaluate our metrics.

3) *Fair Edge Dropouts via Dyadic Masking*: In this method edges are dropped similar to the method above, however for the selected dyads in the population of Sub-Group dyads as see in Algorithm 2 The probability is shifted or biased based on a parameter δ to drop edges with a lot higher probability compared to masked edges in the non-selected Sub-Group dyads. This is adopted from [17]. In our experiments we shift report a combined probability composing of the base probability $p = 0.5$ and δ . Thus, a value of 0.65 suggests an bias of $\delta = 0.15$ towards the selected groups.

G. Post-processing Method

In order to compare and contrast the adopted pre-processing methods we also implemented a simple post-processing method that poses reducing the fairness metrics as an multi-objective optimization problem using a simple SciPy LBFGS-B optimizer [4]. This is done by using the final classification layer's predictions p , the ground-truth labels y and the dyadic groups d without altering the embeddings itself as see in Algorithm 3. With a constraint of $w \geq 0$ since the new-weights w are used for re-ranking each dyadic-group. [12] [15]

IV. RESULT & DISCUSSION

Here we present the results of the experiments and discuss the main takeaways.

A. GCN and GAT behaviour

GCN and GAT show results which are consistent with their different aggregation schemes. It is observed that the behaviour of our GNN's w.r.t. fairness depends on its aggregation function as they perform slight differently with

TABLE III: Effect of layer-depth on GAT & GCN architectures.

Layers	Loss	Acc $\uparrow$	AUC $\uparrow$	F1 $\uparrow$	Pr	Rec	MAP	$\Delta SP_s \downarrow$	$\Delta EO_s \downarrow$	RB _{Node} $\downarrow$
GAT _{L2}	0.3840	0.7952	0.8727	0.6911	0.6874	0.6949	0.5783	0.7692 (0.3170)	1.0000 (0.422, 0.1351)	0.487
GAT _{L4}	0.4489	0.7894	0.8674	0.6617	0.6180	0.7120	0.5234	0.6000 (0.2616)	0.8571 (0.5415, 0.1016)	-
GCN _{L2}	0.4522	0.7609	0.8296	0.6589	0.6928	0.6282	0.5719	0.7846 (0.3523)	1.0000 (0.6322, 0.2044)	0.52
GCN _{L4}	0.4664	0.8006	0.9143	0.7415	0.8582	0.6527	0.7124	0.9846 (0.3058)	1.0000 (0.8065, 0.2530)	0.4887

TABLE IV: Fairness measures after post-processing using greedy Dyadic Fairness optimization.

Method	Acc $\uparrow$	AUC $\uparrow$	F1 $\uparrow$	Pr	Rec	MAP	$\Delta SP_s \downarrow$	$\Delta EO_s \downarrow$
GAT + Dyadic_Post	0.7840	0.8633	0.6555	0.6874	0.6998	0.5195	0.5385 (0.2863)	1.0000 (0.5901, 0.1109)
GCN + Dyadic_Post	0.7617	0.8287	0.6628	0.7027	0.6272	0.5800	0.7846 (0.3596)	1.0000 (0.6520, 0.2078)

Algorithm 2 FairDrop with Dyadic Sub-Group Filtering

Input: Adjacency matrix A , Edge set E , User attributes $\mathbf{u}$, Movie attributes $\mathbf{m}$, Dyadic groups D , Set of top-3 dyadic sub-groups D_3 , Base dropout probability p , Probability adjustment δ , Random distribution $\mathcal{P}$

Output: Fair adjacency matrix A_{fair}

```

1: for all  $e_{i,j} \in E$  do
2:    $d_{i,j} \leftarrow (\mathbf{u}_i, \mathbf{m}_j)$   $\triangleright$  Define dyadic sub-group
3:    $\mathcal{M}_{i,j} = 1$  if  $d_{i,j} \in D_3$ , else 0  $\triangleright$  Construct dyadic mask
4: end for
5: for all  $e_{i,j} \in E$  do
6:    $q_{i,j} \leftarrow p + \delta \mathcal{M}_{i,j}$   $\triangleright$  Adjust dropout probability
7:    $r_{i,j} \sim \mathcal{P}(q_{i,j})$   $\triangleright$  Sample retention probability
8:    $\mathcal{M}_{i,j} = 1$  if  $r_{i,j} \leq q_{i,j}$ , else 0  $\triangleright$  Update mask based on retention
9: end for
10:  $A_{\text{fair}} \leftarrow A \circ \mathcal{M}$   $\triangleright$  Apply mask to adjacency matrix
11: return  $A_{\text{fair}}$ 

```

Algorithm 3 Optimization for Dyadic Sub-Group Fairness

Input: Probabilities $\mathbf{p}$, Ground truth $\mathbf{y}$, Dyadic group indices $\mathbf{d}$, Hyperparameters $\lambda_1, \lambda_2, \lambda_3$

Output: Optimized weights $\mathbf{w}$

```

1:  $\mathbf{w} \leftarrow \mathbf{1}$   $\triangleright$  Initialize weights
2:  $\mathcal{L}(\mathbf{w}) = \lambda_1 \cdot \Delta SP(\hat{\mathbf{y}}, \mathbf{y}, \mathbf{d}) + \lambda_2 \cdot \Delta EO(\hat{\mathbf{y}}, \mathbf{y}, \mathbf{d}) - \lambda_3 \cdot \text{AUC}(\hat{\mathbf{y}}, \mathbf{y})$   $\triangleright$  Combine metrics into the optimization objective
3: Solve  $\mathbf{w}^* = \arg \min_{\mathbf{w} \geq 0} \mathcal{L}(\mathbf{w})$   $\triangleright$  Minimize loss using L-BFGS-B
4: for all  $d_i \in D$  do
5:    $\hat{\mathbf{p}}_{d_i} \leftarrow w_{d_i} \cdot \mathbf{p}_{d_i}$   $\triangleright$  Reweight probabilities for each dyadic group
6: end for
7:  $\hat{\mathbf{y}}_i = 1$  if  $\hat{\mathbf{p}}_i > 0.5$ , else 0  $\triangleright$  Generate binary predictions from reweighted probabilities
8: return  $\mathbf{w}^*$ 

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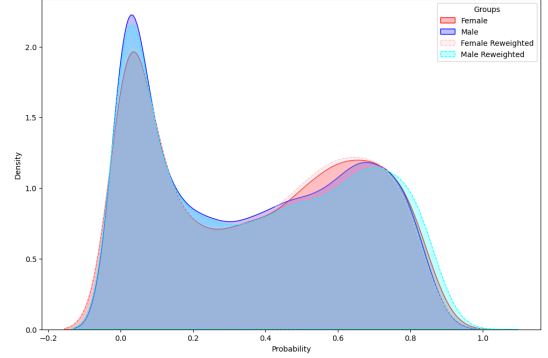


Fig. 2: Re-weighted probability distribution of male and female GCN predictions after post-processing

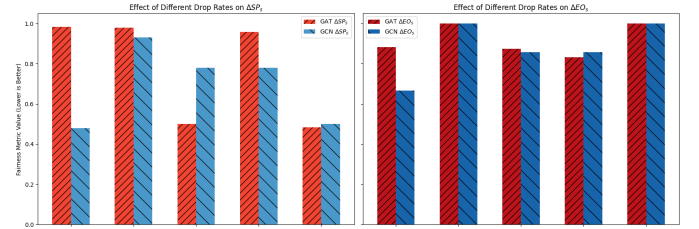


Fig. 3: Dyadic Fairness metric(SP right, EO left) for GAT and GCN for different edge-dropout rates.

addition of layers. It is noticed since GAT uses attention it is able to capture more specialized information in the node-embeddings and achieve good performance with less layers. An explanation of the improvement in fairness as more layers are added could be due to oversmoothing effect [16], since representations would be more similar. For GCN's due to its simple aggregation scheme the fairness metrics are good but more layers are required to account for performance by learning more specialized information.

As observed in Table III GCN requires more hops to capture more specialized relations due to lack of attention based mechanism while aggregating from neighbors. GAT & GCN seem to have an almost opposite behavior w.r.t fairness metrics. Where GAT showcases better fairness on deeper layers, GCN shows the opposite trend with shallower networks performing better.

TABLE V: Performance measures for Random Edge Drop (RD) and FairDrop (FD) on GAT and GCN.

Method	Loss	Acc $\uparrow$	AUC $\uparrow$	F1 $\uparrow$	Pr	Rec	MAP	$\Delta SP_s \downarrow$	$\Delta EO_s \downarrow$	$RB_{Node} \downarrow$
GAT + RD	0.3873	0.7979	0.8689	0.6960	0.6941	0.6980	0.5846	0.9831 (0.3060)	0.8824 (0.5944, 0.1311)	0.5056
GAT + FD _{0.55}	0.3878	0.7896	0.8672	0.6578	0.6066	0.7184	0.5151	0.9796 (0.3019)	1.0000 (0.6275, 0.1777)	0.5147
GAT + FD _{0.6}	0.4053	0.7694	0.8548	0.6079	0.5364	0.7015	0.4523	0.5000 (0.2334)	0.8750 (0.4830, 0.0989)	0.483
GAT + FD _{0.65}	0.3997	0.7743	0.8548	0.6214	0.5556	0.7049	0.4692	0.9583 (0.3058)	0.8333 (0.5851, 0.1311)	0.5105
GAT + FD _{0.7}	0.4014	0.7776	0.8540	0.6489	0.6168	0.6846	0.5170	0.4831 (0.2941)	1.0000 (0.6145, 0.1448)	0.496
GCN + RD	0.4728	0.7074	0.7803	0.4934	0.4274	0.5835	0.3511	0.4787 (0.2414)	0.6667 (0.3906, 0.1415)	0.52
GCN + FD _{0.55}	0.4529	0.7451	0.8136	0.6605	0.7438	0.59	0.6113	0.9310 (0.4055)	1.0000 (0.7263, 0.3185)	0.4957
GCN + FD _{0.6}	0.4646	0.7633	0.8317	0.6672	0.7118	0.6278	0.5875	0.8571 (0.3661)	0.8750 (0.6634, 0.3165)	0.477
GCN + FD _{0.65}	0.4646	0.7592	0.8265	0.6356	0.6299	0.6414	0.5214	0.7808 (0.3232)	0.8571 (0.5697, 0.2946)	0.5
GCN + FD _{0.7}	0.3617	0.8172	0.8847	0.7065	0.6600	0.7601	0.5711	0.5000 (0.2727)	1.0000 (0.6016, 0.1177)	0.507

B. FairDrop improves ΔEO_s

FairDrop Algorithm 2 showed to improve the Equal of Opportunity score. Where the post-processing methods struggles heavily with optimizing EO the adopted FairDrop method seems to somewhat decrease EO scores when $\lambda_1 = \lambda_2$ is set to 1 and $\lambda_3 = 0.7$. As observed by Figure 2 there is only a minimal change in probability distributions w.r.t to re-adjustment of the peaks and slight shift in the distribution of males. We think this is because post-processing methods a need more well-crafted objective, and maybe multiple cascaded optimization steps since balancing both AUC and ΔSP_s , ΔEO_s with only final softmax predictions makes for a harder optimization process.

C. Absence of Representational Bias in User nodes

A surprising result of our experiments is the apparent absence of bias w.r.t to gender when using only user representation. This means that the user nodes themselves do not contain gender information. This would suggest that one cannot assume causal relationships between user nodes and gender bias. As seen by our results the Node Representational Bias scores are always around $\approx 50\%$. A combination of source and target nodes would be required for properly measuring representational bias in heterogenous bipartite networks.

D. Need for better metrics than Parity & Equality Metrics

Our results suggest that the EO and SP metrics used are not adapted to a recommender system. As we see from the TPR and FPR matrices obtained from our fairness metrics as in Fig. 4a and Fig. 4d, we notice that in a dyadic scenario, where fairness depends not just on the source features but also the target features distribution, measuring fairness might be difficult. This is because the metrics are framed in a manner that only consider the difference between the maximum and minimum values across groups. As a result the bias measured shifts from the gender's themselves to that of dyadic groups. This results in an effect where similar ΔEO_s and ΔSP_s values exhibit very different distributions across dyadic groups. Furthermore, bias is measured across all dyadic group, but one could argue that there exist a good bias, used to provide personalized recommendation, and undesired bias, meaning that the recommendations are biased towards the most represented category of user in the training dataset. This is phenomena is also something that was observed by [11] and [6] where

they discuss effect of differing base-rates on group fairness metrics. In the recommendation problem, this would mean negating effect of gender totally, thereby reducing any effect of personalization. This can be seen in our figures that most of the TPR and FPR rates are almost uniform as indicated by the light colours for cases where both ΔEO_s and ΔSP_s are optimized. For the opposite case Fig. 4b, 4c where both metrics are higher we notice all matrices are uniform but darker in colour. In our experiments we noticed that including the sensitive attributes leads to better convergence and improvement of accuracy as well. As discussed in [14] maintaining the effect of gender might not be bad itself. But using metrics such as above may totally eradicate good biases.

Limitations:

The results of this work should be taken critically since the results are reported for only a single dataset, as well for a single seed due to lack of compute hence deviations are not reported. Another limitation is with respect to untested residual connections which could theoretically improve the oversmoothing problem, however has been excluded due to shallow network depth. The post-processing method implemented suffers due to the bi-modal nature of the distribution due to binarization of ratings. A method which accounts for the bi-modal distribution and introduces additional smoothing parameters to discount the contribution between original and new-distributions could potentially reduce ΔEO_s . Adopting FairDrop to such topologies may require a better heuristic. Bipartite structures lack nodes with connections with similar sensitive attributes in the same set. This could be utilized in the adoption to consider not only Edge Dropouts but also Edge additions. As seen in our results surprisingly the measurements for node-representational bias are inconclusive and future experiments should consider whether movie node representations are able to infer gender and if the combination of the movie and user representations possess a bias.

V. REFLECTION & INTUITION DEVELOPMENT

An interesting observation is that accuracy and fairness always comes at a trade-off. Therefore, irrespective of the architecture we chose or de-biasing method we involve there will always be some amount of bias. As we see in our results a system that recommends movies almost similarly across groups suffers somewhat in accuracy. The major flaw with most de-biasing techniques is that we somewhat want

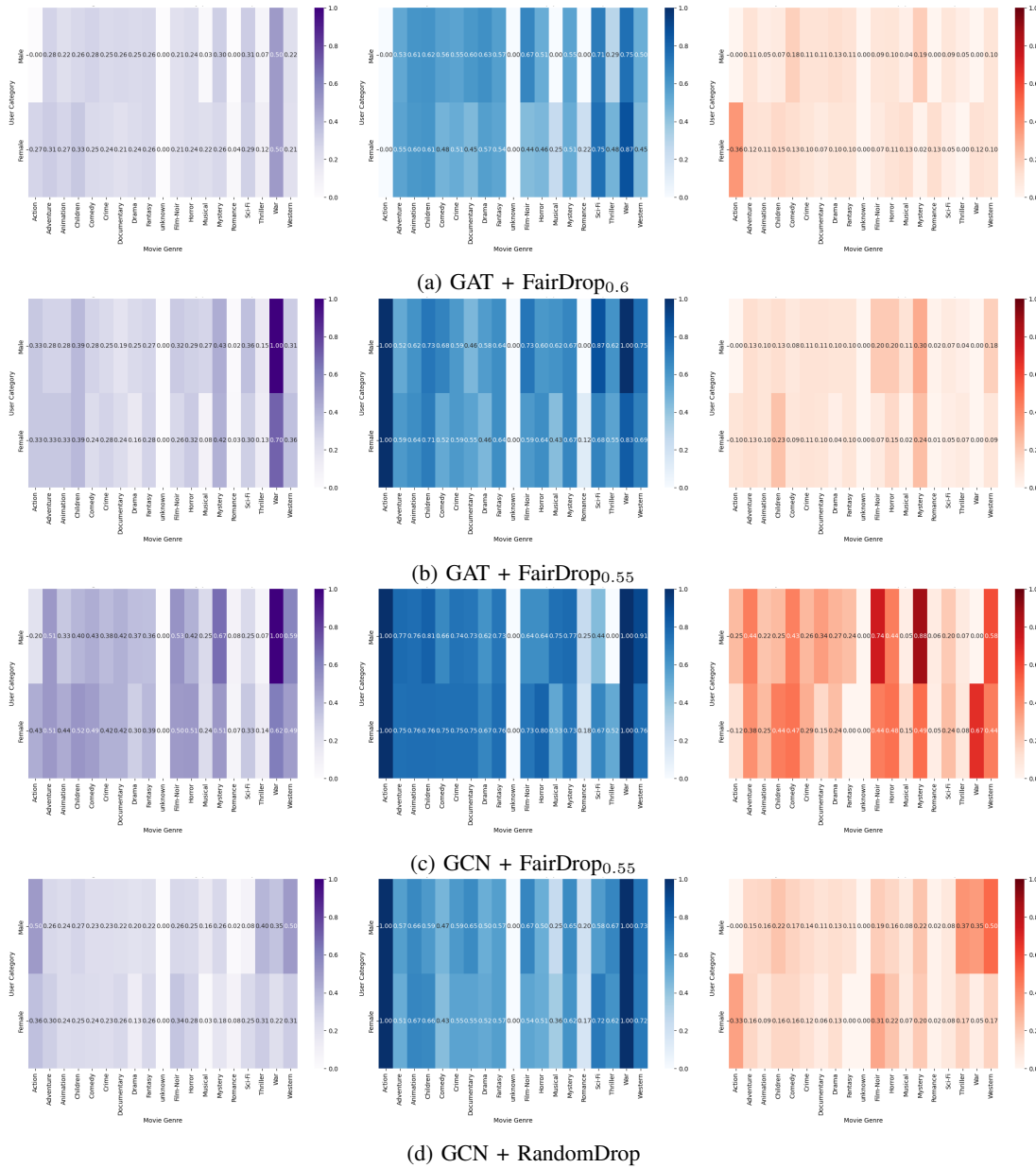


Fig. 4: Avg. link probability, TPR, FPR matrices for different groups used for SP & EO calculations for various models. We observe the spike in avg link probability values and an almost opposite effect of Precision and Recall for GCN and GAT.

to eliminate the impact of the interaction of the sensitive gender features when making predictions. This is not really ideal, as one could argue in essence that then we shouldn't include such attributes altogether. However, the issue is much deeper since we realize after our experiments that the sensitive features of age may have some correlation to other features in the dataset which is true but at the same time we want the good property of this feature. This means the feature itself is not bad but the correlation and interaction causes the bias! Removing such interaction effects are not handled through the method I propose. The method I propose is just a starting. Given we have proposed a Fair recommender system, its important to realise what fairness would even mean? If we

use a naive method as I proposed this would mean somewhat that both gender groups receive a small percentage of random recommendations. One may also argue that there is not a need for fairness altogether, as users may prefer to watch the movies based on their gender characteristics. That is personalization effect could be preferred over a fair recommendation system as it may lead to reduced engagement and dissatisfaction. Therefore, this method explores an ideal scenario where gender characteristics may not influence the movie watching dynamics themselves. More research is required into whether the fairness approaches impact the satisfaction and engagement for our users which this paper doesn't tackle.

VI. CONCLUSION

We compare how using a simple edge-based dyadic pre-processing method adopted from FairDrop improves sub-group fairness compared to random Edge Dropouts and greedy post-processing in heterogeneous bipartite networks at a slight trade-off for accuracy. Our results showcase that the fairness metrics may mask the variation between groups when comparing different GNN models and motivate the use of others metrics for heterogeneous bi-partite graph.

REFERENCES

- [1] Alekh Agarwal et al. “A reductions approach to fair classification”. In: *International conference on machine learning*. PMLR. 2018, pp. 60–69.
- [2] Maarten Buyt and Tijl De Bie. “Debayes: a bayesian method for debiasing network embeddings”. In: *International Conference on Machine Learning*. PMLR. 2020, pp. 1220–1229.
- [3] Maarten Buyt and Tijl De Bie. “The kl-divergence between a graph model and its fair i-projection as a fairness regularizer”. In: *Machine Learning and Knowledge Discovery in Databases. Research Track: European Conference, ECML PKDD 2021, Bilbao, Spain, September 13–17, 2021, Proceedings, Part II 21*. Springer. 2021, pp. 351–366.
- [4] Richard H Byrd et al. “A limited memory algorithm for bound constrained optimization”. In: *SIAM Journal on Scientific Computing* 16.5 (1995), pp. 1190–1208. DOI: 10.1137/0916069.
- [5] April Chen et al. “Fairness-aware graph neural networks: A survey”. In: *ACM Transactions on Knowledge Discovery from Data* 18.6 (2024), pp. 1–23.
- [6] Alexandra Chouldechova. “Fair prediction with disparate impact: A study of bias in recidivism prediction instruments”. In: *Big data* 5.2 (2017), pp. 153–163.
- [7] Yushun Dong et al. “Fairness in graph mining: A survey”. In: *IEEE Transactions on Knowledge and Data Engineering* 35.10 (2023), pp. 10583–10602.
- [8] Sorelle A Friedler, Carlos Scheidegger, and Suresh Venkatasubramanian. “The (im) possibility of fairness: Different value systems require different mechanisms for fair decision making”. In: *Communications of the ACM* 64.4 (2021), pp. 136–143.
- [9] Moritz Hardt, Eric Price, and Nati Srebro. “Equality of opportunity in supervised learning”. In: *Advances in neural information processing systems* 29 (2016).
- [10] Thomas N. Kipf and Max Welling. *Semi-Supervised Classification with Graph Convolutional Networks*. 2017. arXiv: 1609.02907 [cs.LG]. URL: <https://arxiv.org/abs/1609.02907>.
- [11] Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. “Inherent trade-offs in the fair determination of risk scores”. In: *arXiv preprint arXiv:1609.05807* (2016).
- [12] Ivona Krchova, Michael Platzer, and Paul Tiwald. “Strong statistical parity through fair synthetic data”. In: *arXiv preprint arXiv:2311.03000* (2023).
- [13] Charlotte Laclau, Christine Largeron, and Manvi Choudhary. “A survey on fairness for machine learning on graphs”. In: *arXiv preprint arXiv:2205.05396* (2022).
- [14] Jing Ma et al. “Learning fair node representations with graph counterfactual fairness”. In: *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*. 2022, pp. 695–703.
- [15] Farzan Masrour et al. “Bursting the filter bubble: Fairness-aware network link prediction”. In: *Proceedings of the AAAI conference on artificial intelligence*. Vol. 34. 01. 2020, pp. 841–848.
- [16] Kenta Oono and Taiji Suzuki. “Graph neural networks exponentially lose expressive power for node classification”. In: *arXiv preprint arXiv:1905.10947* (2019).
- [17] Indro Spinelli et al. “Fairdrop: Biased edge dropout for enhancing fairness in graph representation learning”. In: *IEEE Transactions on Artificial Intelligence* 3.3 (2021), pp. 344–354.
- [18] Petar Veličković et al. *Graph Attention Networks*. 2018. arXiv: 1710.10903 [stat.ML]. URL: <https://arxiv.org/abs/1710.10903>.
- [19] Todd Wasserman. *What If Online Movie Ratings Weren’t Based Almost Entirely on What Men Think?* Accessed: 2024-10-14. 2018. URL: <https://fivethirtyeight.com/features/what-if-online-movie-ratings-werent-based-almost-entirely-on-what-men-think/>.
- [20] Zonghan Wu et al. “A comprehensive survey on graph neural networks”. In: *IEEE transactions on neural networks and learning systems* 32.1 (2020), pp. 4–24.
- [21] Rex Ying et al. “Graph convolutional neural networks for web-scale recommender systems”. In: *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining*. 2018, pp. 974–983.
- [22] Ziqian Zeng et al. “Fair representation learning for heterogeneous information networks”. In: *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 15. 2021, pp. 877–887.